

The Universality of Electricity Demand: Zero-Shot Foundation Models Match Locally-Trained Accuracy on Brazil’s Power Grid

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Abstract—Short-term load forecasting (STLF) has traditionally required models trained on years of local historical demand—a barrier in data-scarce grids. We ask whether recent time series *foundation models*, applied zero-shot, can forecast Brazilian electricity load with no local training. Using seven years (2019–2025) of hourly ONS load across all four subsystems of the interconnected grid (SIN), we benchmark Chronos-2 (120 M), TiRex (35 M), and Moirai 2.0 (11 M) against a hyperparameter-tuned N-BEATS, a linear model, an LSTM, and a seasonal-naïve baseline. On the Southeast/Center-West (SE) subsystem at a 24-hour horizon, zero-shot Chronos-2 attains 1.86 % MAPE—statistically indistinguishable from a tuned N-BEATS trained on 5+ years of local data ($1.91\% \pm 0.08\%$; Diebold–Mariano $p > 0.29$). Light fine-tuning yields only a 7 % relative gain (1.73 %). The result holds across all four subsystems (1.67–3.17 % MAPE, $R^2 > 0.90$), three test years ($1.89\% \pm 0.04\%$), and—on the Tokyo (TEPCO) grid, a different country and hemisphere—where the same three zero-shot models again beat a locally-trained N-BEATS (3.91 % vs. 4.44 %). The dominant residual error concentrates on national holidays (4–5× the normal-day error); adding a single binary holiday covariate cuts holiday error by 35 %. We read this as evidence for a *bounded universality* of electricity demand: physics- and behavior-driven structure transfers across grids and hemispheres without adaptation, while calendar-specific demand does not.

Index Terms—Short-term load forecasting, foundation models, zero-shot learning, power systems, Chronos, Moirai, TiRex, N-BEATS.

I. INTRODUCTION

Short-term load forecasting (STLF) underpins unit commitment, economic dispatch, reserve sizing, and day-ahead market clearing. The dominant paradigm has each utility build a bespoke model trained on years of local load, augmented with weather and calendar features. This imposes a heavy barrier in data-scarce contexts—newly interconnected regions, microgrids, and developing grids lacking long, clean records.

Time series foundation models (TSFMs)—large networks pretrained on billions of observations across diverse domains—raise a sharp question: can a model that has never seen a particular grid’s data forecast its load accurately? If so, forecasting could be deployed without the data-collection and maintenance overhead of local training.

We frame this as a hypothesis of *universality*. Demand is shaped by two kinds of forces. The first are universal—the physics of heating, cooling, and lighting; the circadian rhythm of activity; the weekly work/rest cycle—producing daily peaks, overnight troughs, and weekend dips in every electrified society. The second are local—national holidays, hydro-dependent supply constraints, and southern-hemisphere seasonality inverted from the northern-hemisphere data these models consume. If a pretrained model has internalized the universal component, it should transfer to an unseen grid zero-shot, with residual error confined to the local component.

We test this primarily on the Brazilian interconnected system (SIN) using public ONS data, with a cross-country replication on the Tokyo (TEPCO) grid. Our contributions are: (1) a systematic benchmark of 2024-era TSFMs on Brazilian load (all four SIN subsystems, four horizons, three test years) plus a cross-country replication on a Japanese grid; (2) evidence that zero-shot models match or beat a tuned, locally-trained N-BEATS (Diebold–Mariano $p > 0.29$ on SE); (3) a precise account of *when and why* they fail—and a holiday covariate that recovers most of the gap; and (4) a reproducible benchmark.

II. RELATED WORK

Brazilian STLF work relies on locally-trained SVR, ANN, ARIMA, and LSTM models on ONS or utility data [6], [7], [8], all requiring per-grid development. The TSFM paradigm emerged in 2023–2024—pretrain on billions of time points, then forecast unseen series zero-shot—with models including Chronos [1], TimesFM [3], Moirai [4], and TiRex [5]. TSFMs have been evaluated on US grids (ERCOT [9]), Singapore/Australia [10], and European households [11], but not on emerging-market grids with hydro dependency, inverted seasonality, and distinct holiday calendars, nor against a locally-trained deep-learning baseline on the same data. We fill that gap.

III. DATA AND METHODOLOGY

Data. Hourly load from the ONS open-data portal (dados.ons.org.br, CC-BY 4.0) for the four SIN subsystems

TABLE I
SIN SUBSYSTEMS AND DATASET STATISTICS (2019–2025)

Sub.	Region	Mean (MW)	Test (MW)	Share
SE	SP, RJ, MG	40,408	44,226	~55 %
S	PR, SC, RS	12,320	13,804	~17 %
NE	BA, PE, CE	11,711	13,267	~17 %
N	AM, PA	6,641	8,312	~11 %

(Table I); SE is the largest at ~55 % of national load (which averages 71,081 MW). Each series spans 2019-01-01 to 2025-12-31 (61,368 hourly rows). The most recent 365 days (8,760 h) form the test set: foundation models see only the preceding context window (zero-shot, never trained on ONS data); trained baselines use all earlier data, reserving the prior 60 days for validation.

Task. Given a context window of hourly load, predict the next 24 h (day-ahead); we also evaluate 168/336/720 h. Forecasts roll through the test year in 24-h steps and are averaged.

Models. Zero-shot: *Chronos-2* (120 M, encoder-only, quantile output; median reported), *TiRex* (35 M, xLSTM), *Moirai 2.0* (11 M). We verified ONS data is *absent* from Chronos’s pretraining corpus [1], [2] (no data leakage). Trained: *N-BEATS* (7.3 M, Darts; primary baseline), *Linear* (~8 K), *LSTM*, and *Naive* (7-day-ago). Metrics: MAE, RMSE (MW), MAPE, MASE/RMSSE (seasonality = 24), R^2 ; CRPS and prediction intervals for probabilistic eval; Diebold–Mariano (DM) test on per-window errors. Hardware: Mac Mini M4, no GPU; MPS fails for Darts N-BEATS and Chronos fine-tuning, run on CPU.

IV. RESULTS

A. Main Comparison (SE, 24 h)

Table II and Fig. 1 report the headline. Zero-shot Chronos-2 achieves **1.86 % MAPE**, edging out a tuned N-BEATS (1.91 %) trained on five years of local data, with Moirai close behind (1.93 %); all TSFMs beat naive (5.13 %) by a wide margin. The DM test gives $p > 0.29$ against all three N-BEATS seeds and bootstrap 95 % CIs overlap (Chronos-2 CI [1.70%, 2.04%]): Chronos-2 vs. N-BEATS is a *statistical tie*—but Chronos-2 needs *zero* local training. Light fine-tuning (400 steps, lr 1e-5, ~40 min CPU) reaches 1.73 % (a 7 % gain), showing pretraining already captures the dominant patterns. The LSTM failed to train competitively and is shown for completeness. For context, Chronos-2 (load only) rivals proprietary ISO systems using weather/calendar inputs: PJM 24 h (1.78–1.98 %), ERCOT (1.66–3.73 %).

N-BEATS tuning. To ensure fairness, N-BEATS was grid-searched over input length $\in \{168, 336, 720\}$ h and lr $\in \{1e-4, 5e-4, 1e-3\}$. The best config (168 h, lr 5e-4), re-run with seeds {42, 123, 7}, gives 1.85/1.88/2.00 % $\rightarrow$ **1.91 % $\pm$ 0.08 %**—a 0.23 pp improvement over its untuned setting (2.14 %), confirming the baseline is properly optimized.

TABLE II
MAIN RESULTS — SE, 24 h HORIZON

Model	Type	MAPE %	MASE	R^2
Chronos-2	Fine-tuned	1.73	0.30	0.96
Chronos-2	Zero-shot	1.86	0.33	0.96
N-BEATS (tuned)	Trained 5 yr	1.91 ± 0.08	0.34	0.96
Moirai 2.0	Zero-shot	1.93	0.34	0.95
Linear	Trained	2.26	0.40	0.94
TiRex	Zero-shot	2.33	0.40	0.94
Naive (7 d)	Baseline	5.13	0.89	0.77
LSTM	Trained	13.14	2.39	−0.36

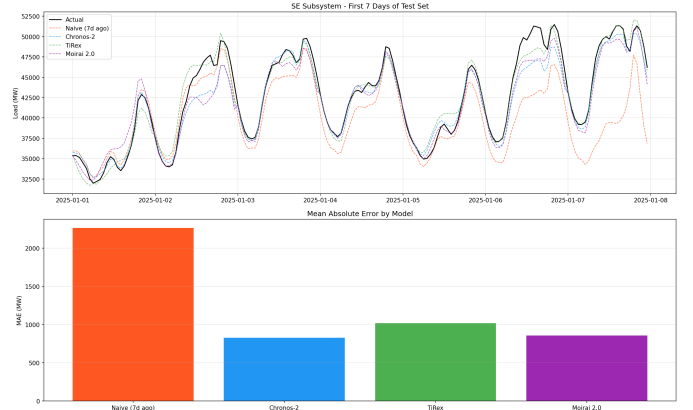


Fig. 1. SE, 24h: 7-day forecast comparison and per-model MAE. Zero-shot Chronos-2 tracks the load curve closely; the naive baseline lags on the daily ramp.

B. Generalization

Across subsystems. Chronos-2 leads on all four (Table III), 45–64 % above naive, $R^2 > 0.90$ everywhere. The smallest subsystem (N) is easiest (1.67 %); S is hardest (3.17 %), likely from winter cold-front heating. On S, tuned N-BEATS reaches $3.26\% \pm 0.21\%$ vs. 3.17 %—again a tie.

Horizon. Accuracy degrades with horizon, with a *naive crossover* at ~2–3 weeks. SE Chronos-2 MAPE: 1.86 % (24 h), 3.59 % (168 h), 4.18 % (336 h), 5.69 % (720 h); at 720 h Chronos-2 (MASE 1.02) and Moirai (1.03) fall behind naive while only TiRex (0.91) stays ahead. TSFMs are thus best for operational (24–168 h) horizons.

Multi-year. Chronos-2 is stable across test years 2023/2024/2025 (1.94/1.87/1.86 %, mean $1.89\% \pm 0.04\%$) while naive varies more ($5.31\% \pm 0.17\%$).

Context length. The critical threshold is *one week*: MAPE halves ($5.65\% \rightarrow 2.80\%$) once a full weekly cycle is seen, reaching 1.86 % at 30 days; beyond that, returns are marginal (90 days: 1.77 %). Thirty days is the practical accuracy/compute sweet spot.

Model scale. The Chronos-Bolt family scales log-linearly (Fig. 2): Tiny 9 M $\rightarrow$ 2.28 %, Mini 21 M $\rightarrow$ 2.19 %, Small 48 M $\rightarrow$ 2.02 %, Chronos-2 120 M $\rightarrow$ 1.86 %. At matched size zero-shot Bolt-Tiny (2.28 %) loses to tuned N-BEATS (1.91 %), but Moirai (11 M, 1.93 %) nearly ties it—architecture matters as much as scale.

TABLE III
MAPE (%) BY SUBSYSTEM — 24 H

Sub.	Naive	Chronos-2	TiRex	Moirai
SE	5.13	1.86	2.33	1.93
S	7.11	3.17	3.37	3.35
NE	3.76	1.94	2.06	2.05
N	3.03	1.67	1.67	1.76

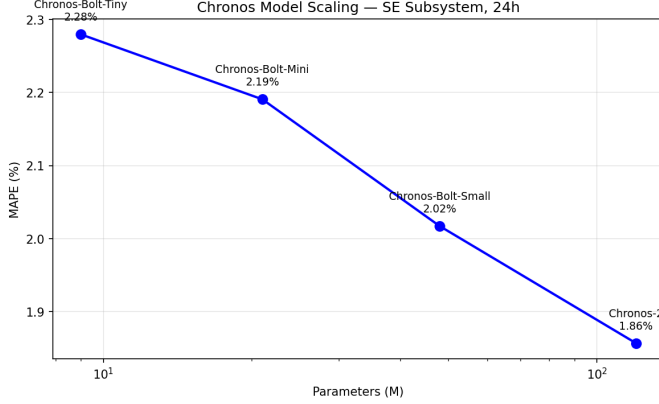


Fig. 2. Chronos scaling: MAPE decreases log-linearly with parameter count; at ~ 9 M zero-shot trails tuned N-BEATS, and by 120 M it closes the gap.

Probabilistic. Both probabilistic models are well-calibrated but slightly conservative—80 % PIs cover 86–89 % of outcomes, desirable for reserve scheduling. Chronos-2 has the better CRPS (643 vs. 688 MW) and Winkler score (4354 vs. 4378); Moirai produces sharper intervals (3024 vs. 3295 MW) despite being $11\times$ smaller.

C. Cross-Country Replication (Tokyo)

To probe universality beyond Brazil, we repeat the benchmark on the TEPCO grid (Tokyo, ~ 32 GW, 2019–2024), with N-BEATS trained on 2019–2023 Japanese data and 2024 held out (Table IV). All three zero-shot foundation models again *beat* the locally-trained N-BEATS, in the identical ranking observed for Brazil (Chronos-2 > Moirai > TiRex $\gg$ naive). Absolute MAPE is higher (Japanese demand is more volatile), but the qualitative result—zero-shot transfer matches or exceeds dedicated local training—holds across a different country, hemisphere, and holiday calendar. This is the strongest single piece of evidence for the universality hypothesis.

V. ERROR ANALYSIS: WHEN AND WHY

Error follows the load curve (Fig. 3): lowest in the overnight trough (0.51 % at hour 0), highest mid-afternoon (2.72 % at hour 15). Weekends (1.75 %) beat weekdays (1.90 %); Monday is hardest (2.37 %), Tuesday easiest (1.60 %).

The central finding is the holiday regime (Table V). On Brazilian public holidays Chronos-2 degrades to several times its normal-day error, because nothing in the global corpus signals a Brazilian holiday, so it forecasts ordinary workday demand. The effect is a property of the data, not one model:

TABLE IV
CROSS-COUNTRY: TEPCO TOKYO — 24 H, TEST 2024

Model	Type	MAPE %	MASE	R^2
Chronos-2	Zero-shot	3.91	0.59	0.91
Moirai 2.0	Zero-shot	3.94	0.59	0.91
TiRex	Zero-shot	4.05	0.61	0.91
N-BEATS	Trained 5 yr	4.44	0.67	0.89
Naive (7 d)	Baseline	8.86	1.30	0.65

TABLE V
HOLIDAY COVARIATE EXPERIMENT — CHRONOS-2, SE (TEST 2024)

Regime	Hours	Zero-shot	+ Cov.	Δ %
Normal	5688	1.53	1.53	+0.0
Weekend	2232	1.77	1.74	−2.0
Day-before	264	2.28	2.19	−4.2
Day-after	264	2.29	1.93	−15.8
Carnaval	48	5.68	4.64	−18.3
Holiday	264	6.76	4.37	−35.4
Bridge	24	3.02	6.12	+102.5
Overall	8784	1.82	1.73	−5.0

on the same holidays Moirai reaches 7.9 % and TiRex 9.1 % MAPE, and the naive baseline 12.8 %, against ~ 1.6 % on normal days.

This is precisely where a small dose of local knowledge should help—and it does. Adding a single binary holiday covariate and lightly fine-tuning Chronos-2 (test 2024) cuts holiday MAPE by 35 % (6.76 % $\rightarrow$ 4.37 %), Carnaval by 18 %, and the day after a holiday by 16 %, while normal days are untouched and overall MAPE falls 5 % (Table V). The lone regression—“bridge” days between a holiday and a weekend worsen (3.02 % $\rightarrow$ 6.12 %), on only 24 test hours—marks where the coarse binary encoding is too blunt and a richer calendar feature is needed. This is exactly the bounded-universality prediction: the residual error is local calendar knowledge, and supplying that knowledge recovers most of the gap.

VI. DISCUSSION AND CONCLUSION

The results support a universality of electricity demand *with a precise boundary*. Components driven by shared physical and behavioral forces—diurnal cycles, the workweek, seasonality—transfer to an unseen grid well enough that a zero-shot model ties or beats a locally-trained one across subsystems, years, and—on the Tokyo grid—a second country and hemisphere. Components driven by local culture—national holidays—do not, degrading 4–5 $\times$; but supplying exactly that local knowledge via a binary holiday covariate recovers 35 % of the holiday gap, confirming the boundary is calendar knowledge rather than an architectural limit.

This yields an operational recipe: (1) start zero-shot (Chronos-2, or the 11 M Moirai when resource-constrained) with ~ 30 days of context; (2) fine-tune if resources permit (~ 7 % gain, 40 min CPU); (3) add a binary holiday covariate (cut holiday error 35 % in our experiment); (4) do not use

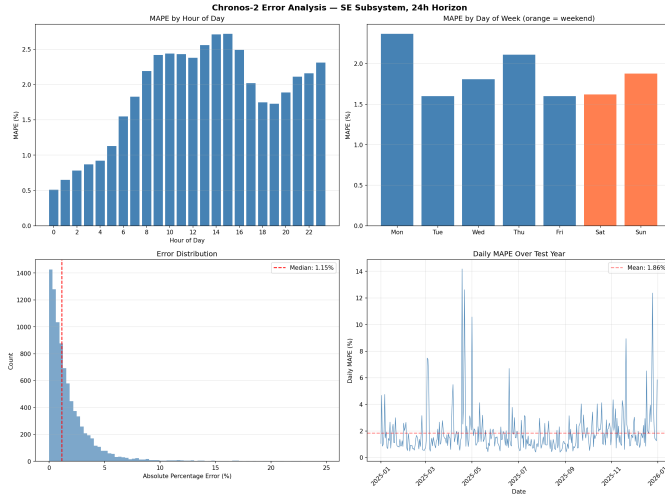


Fig. 3. Error decomposition: MAPE by hour-of-day and day-of-week, error distribution, and daily MAPE over the test year (spikes coincide with holidays).

beyond ~ 2 weeks, where naive wins; (5) one week of history is the minimum viable context. *Limitations*: univariate input (no weather); two countries (Brazil, Japan) remain a small sample of grids; load-only trained baselines, so a weather-augmented TFT could narrow the gap; the holiday covariate’s binary encoding regresses on rare “bridge” days; and the LSTM did not converge competitively.

In sum, a model pretrained on global time series, with no exposure to the target grid’s data, matches a tuned model trained on 5+ years of local Brazilian data (1.86% vs. $1.91\% \pm 0.08\%$; DM $p > 0.29$) across all four subsystems and three years, and *beats* a locally-trained N-BEATS on the Tokyo grid (3.91% vs. 4.44%). The boundary of universality is precise—the model fails on culturally-local holidays and essentially nowhere else—and a single holiday covariate largely closes even that gap. For operators in data-scarce grids, accurate day-ahead forecasting no longer requires years of local model development: thirty days of load history and an off-the-shelf foundation model suffice.

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